THE UNIVERSITY OF AZAD JAMMU AND KASHMIR

Muzaffarabad

Department of Software Engineering

LAB 2

Software Construction & Development (Lab)

Semester: 5th

Course Code: SE-3102

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Question 1: Activity Diagram of foodpanda Using Swimlanes

Objective

To show the online food-order process and separate responsibilities between the customer, foodpanda app, and restaurant/rider.

Swimlanes and Responsibilities

Customer: Opens the app, enters a location, browses restaurants, selects food, confirms or edits the order, pays, and receives delivery.

foodpanda App: Displays menus and prices, calculates the total, records the order, and communicates order status.

Restaurant / Rider: Receives and accepts the order, prepares the food, collects it, and delivers it.

Main Process

Start: customer opens the app and enters a delivery location.

Customer browses restaurants and selects food.

App displays the menu and calculates the total.

Customer confirms the order; otherwise, the customer may edit or cancel it.

Customer chooses a payment method and submits payment.

If payment succeeds, the order is sent to the restaurant; if it fails, the customer can retry or choose another method.

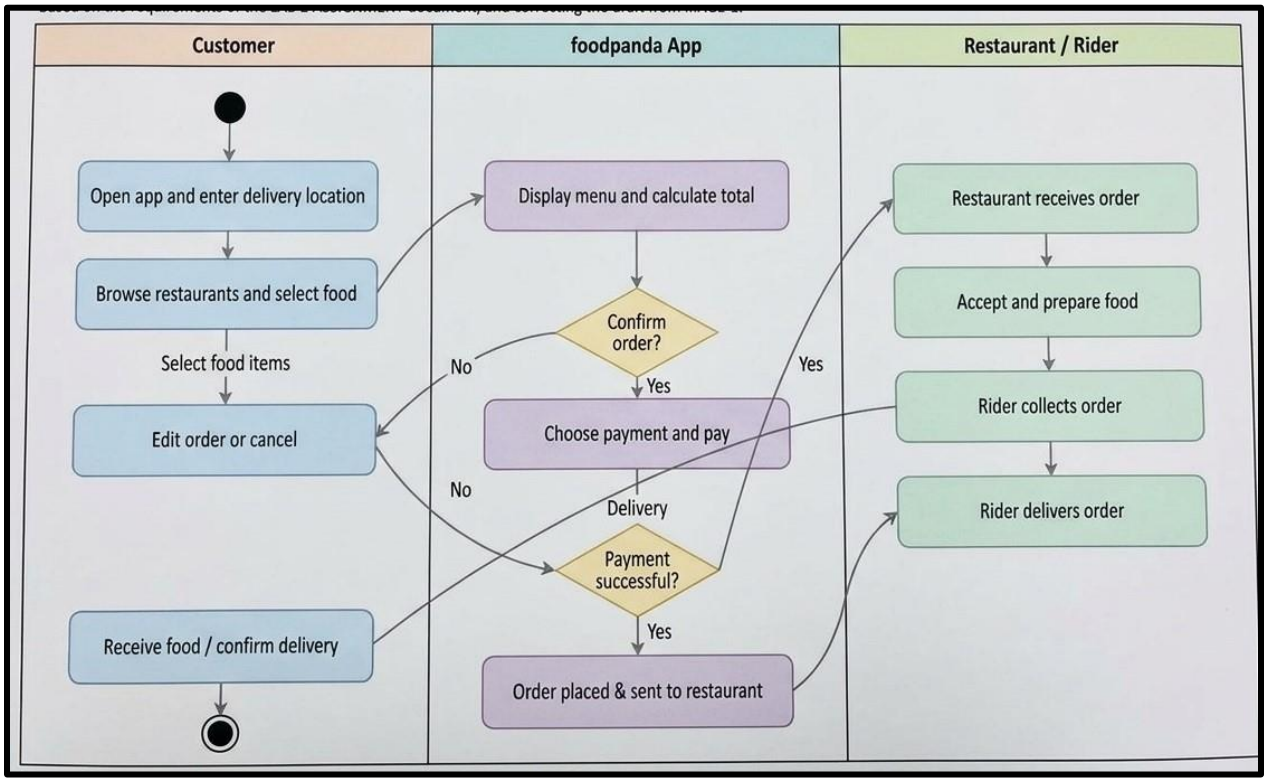
Restaurant accepts and prepares the food.

Rider collects the order and delivers it.

Customer receives the food; the process ends.

Activity Diagram (Swimlane)

This is a simplified educational workflow. Actual app screens and payment options may vary.



Question 2: State Diagram of a Toaster

Objective

To identify the toaster's states, activities within each state, and triggered or triggerless transitions.

States and Activities

Idle / Ready: Wait for bread or lever operation; heating elements remain off.

Toasting: Heat the bread, count down the timer, and monitor the cancel control.

Ejecting: Release the latch and raise the bread carriage.

Finished: Keep the toast raised and wait for it to be removed.

Fault / Cool-down (assumption): Stop heating and cool down if overheating is detected; this safety state is an educational assumption.

Triggered Events and Actions

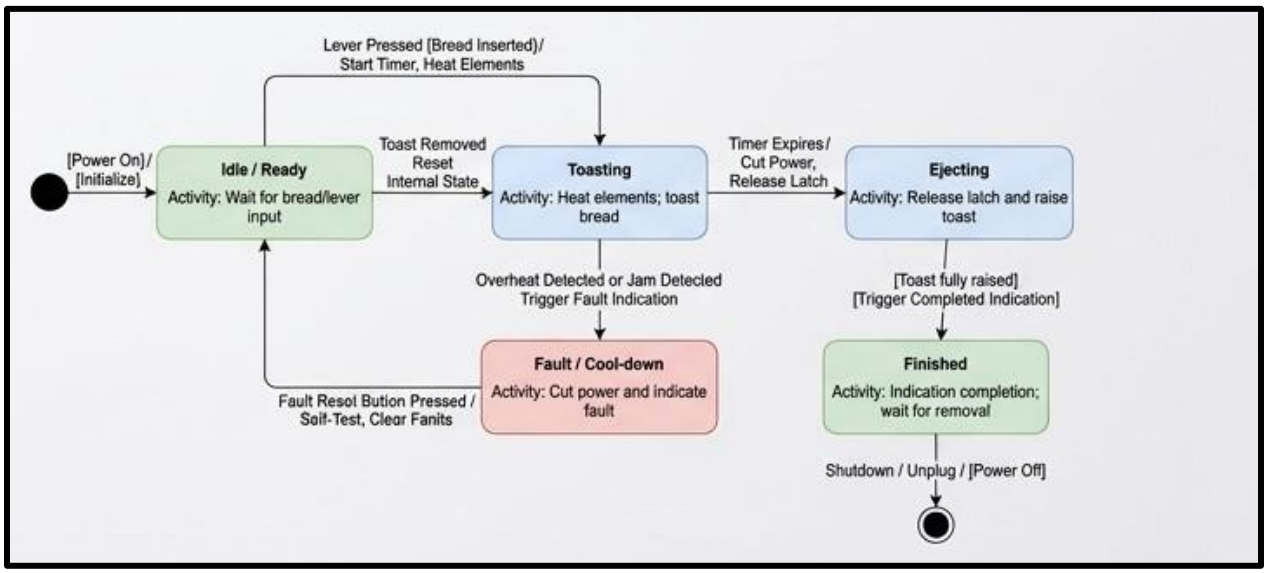
Current State	Event / Trigger	Action / Next State
Idle / Ready	Lever pushed	Start heating → Toasting
Toasting	Timer expires	Cut power → Ejecting
Toasting	Cancel pressed	Stop heating → Ejecting / Idle
Ejecting	Carriage reaches top	Raise toast → Finished
Finished	Toast removed	Reset → Idle / Ready
Any active state	Overheat detected (assumption)	Cut power → Fault / Cool-down
Fault / Cool-down	Safe temperature + reset	Return to Idle / Ready

Triggered vs. Triggerless

Triggered event: a transition starts because an event occurs, such as the lever being pushed, timer expiring, cancel button being pressed, or toast being removed.

Triggerless completion transition: a transition occurs automatically when an activity/state completes, without a new external event. For example, completion of the heating cycle causes power to be cut and the toaster to begin ejecting. In UML, this may be represented as a completion transition.

State Diagram



Question 3: State Machine Diagram of an Elevator / Lift

Objective

To model the lift's main states and transitions in response to requests, door conditions, floor arrival, and safety events.

States and Activities

Idle at Floor: Wait for a request; motor stopped and doors closed.

Doors Opening: Operate the door mechanism to open the doors.

Doors Open: Allow passengers to enter or exit; hold doors if an obstruction is detected.

Doors Closing: Close the doors and check the safety interlock.

Moving: Move toward the requested floor while monitoring position and safety signals.

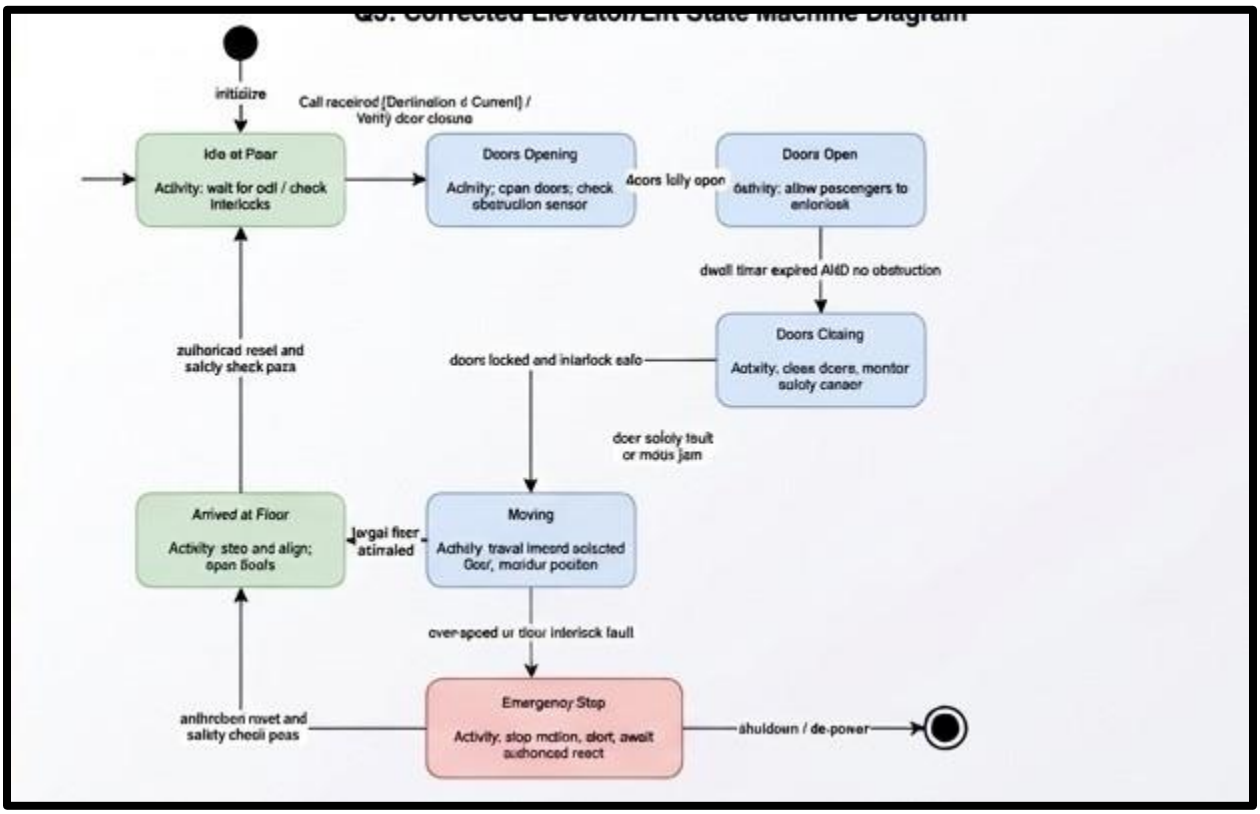
Arrived at Floor: Stop the motor and prepare to open the doors.

Emergency Stop: Stop movement and remain safe until reset and inspection.

Main Transitions

Current State	Event / Condition	Next State / Action
Idle at Floor	Request received	Doors Opening
Doors Opening	Doors fully open	Doors Open
Doors Open	Timeout / close command [no obstruction]	Doors Closing
Doors Closing	Doors closed and interlock safe	Moving
Moving	Target floor reached	Arrived at Floor
Arrived at Floor	Open doors	Doors Opening
Doors Closing / Moving	Emergency or overload signal	Stop / hold safely as appropriate
Emergency Stop	Reset and safety conditions verified	Idle at Floor

State Machine Diagram



Safety note: This is a simplified academic diagram, not a design for controlling real elevator equipment.

Conclusion

The assignment applies UML behavioral diagrams to three systems. The foodpanda activity diagram uses swimlanes to distinguish the customer, app, and restaurant/rider responsibilities. The toaster state diagram identifies states, activities, and event-driven or completion-driven

transitions. The elevator state machine shows normal operating states and simplified safety transitions.

*****THE END*****